

ECOM30001/90001 Basic Econometrics

Tutorial 5: Restricted vs. Unrestricted Models & Ramsey RESET

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Tutorial 5 Overview & Key Concepts

Core Topics

- **Joint Parameter Restrictions:** Deriving, testing, and relaxing linear restrictions on regression coefficients.
- **Compromise Models & Omitted Variable Bias:** Examining partial restriction relaxation and the impact of excluding relevant regressors.
- **Cost Function Modeling:** Testing linear, quadratic, and cubic total cost functions, as well as linear average cost functions.
- **Specification Testing (Ramsey RESET):** Using powers of fitted values ($\hat{Y}^2$, $\hat{Y}^3$) to detect functional form mis-specification.
- **R Packages Practice:** Utilizing 'car', 'lmtest', 'modelsummary', and 'rio'.

Question 1(a)-(b): Deriving Linear Restrictions

Restricted vs. Unrestricted Specifications

Model (1) [Restricted]:

$$\ln \left(\frac{L_t}{K_t} \right) = \beta_0 + \beta_1 \ln \left(\frac{w_t}{r_t} \right) + \beta_2 \ln Y_t + \varepsilon_t$$

Model (2) [Unrestricted]:

$$\ln \left(\frac{L_t}{K_t} \right) = \alpha_0 + \alpha_1 \ln w_t + \alpha_2 \ln r_t + \alpha_3 \ln Y_t + \alpha_4 \ln K_t + \varepsilon_t$$

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Derivation & Imposed Restrictions:

- Expand Model (1):
 $\ln(L_t/K_t) = \beta_0 + \beta_1 \ln w_t - \beta_1 \ln r_t + \beta_2 \ln Y_t + (1 - 1) \ln K_t + \varepsilon_t.$
- Comparing coefficients with Model (2) reveals **two linear restrictions**:

$$\alpha_1 + \alpha_2 = 0 \quad (\text{since } \beta_1 - \beta_1 = 0), \quad \alpha_4 = 0 \quad (\text{since } 1 - 1 = 0)$$

Question 1(c)-(d): Joint Test of Restrictions

Hypothesis Test

$H_0 : \alpha_1 + \alpha_2 = 0 \quad \text{and} \quad \alpha_4 = 0 \quad \text{vs.} \quad H_1 :$
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R Results ('linearHypothesis') & Manual F -Test:

- **R Code:** `'linearHypothesis(eq2, c("log(w)+log(r)=0", "log(K)=0"))'`
- **Residual Sum of Squares:** $RRSS = 0.72838$ (Model 1), $URSS = 0.03707$ (Model 2).

- **F -Statistic Calculation** ($q = 2, n - k - 1 = 27$):

$$F = \frac{(RRSS - URSS)/q}{URSS/(n - k - 1)} = \frac{(0.72838 - 0.03707)/2}{0.03707/27} = 251.73 \sim F_{2,27}$$

- **p -value:** $p = 3.47 \times 10^{-18} < 0.05 \implies$ **Decisively reject H_0 .**

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Conclusion

Imposing both restrictions simultaneously is rejected at the 5% level. Model (1) is overly restrictive relative to Model (2).

Question 1(e)-(f): Compromise Model & OVB

Compromise Model (3)

Imposes *only* $\alpha_1 + \alpha_2 = 0$ while relaxing $\alpha_4 = 0$:

$$\ln \left(\frac{L_t}{K_t} \right) = \gamma_0 + \gamma_1 \ln \left(\frac{w_t}{r_t} \right) + \gamma_2 \ln Y_t + \gamma_3 \ln K_t + \varepsilon_t$$

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Key Insights & Omitted Variable Bias (OVB):

- **Why Model (3) is Best Compromise:** In Model (2), $a_1 + a_2 = 0.0293 - 0.1293 = -0.10$ (close to 0), but $a_4 = -1.075$ (far from 0).
- **Major Parameter Shift:** Excluding $\ln K_t$ in Model (1) flips the sign of $\ln(w/r)$ and $\ln Y$ coefficients compared to Model (3).
- **Takeaway:** Excluding $\ln K_t$ induces severe **omitted variable bias**. Model (3) captures the strong labour-saving effect of capital investment ($\hat{\gamma}_3 < -1$).

Question 2(a)-(b): Testing Linear Total Cost

Cubic Total Cost Model Setup ($n = 28$ clothing firms)

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \beta_3 X_i^3 + \varepsilon_i$$

Where $Y_i = \text{Total Cost}$, $X_i = \text{Output}$, $X_i^2 = \text{Output}^2$, $X_i^3 = \text{Output}^3$.

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Linear Total Cost Test ($H_0 : \beta_2 = \beta_3 = 0$):

- Linear total cost $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ requires $\beta_2 = \beta_3 = 0$.
- R Output ('linearHypothesis')**: `'linearHypothesis(mod, c("X2 = 0", "X3 = 0"))'`

$$F = 62.056, \quad p\text{-value} = 3.28 \times 10^{-10} < 0.05$$

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Conclusion

Since $p < 0.05$, we **reject** H_0 at the 5% significance level. The total cost function is not linear.

Question 2(c)-(d): Quadratic Cost & Linear Average

Part (c): Quadratic Total Cost Test ($H_0 : \beta_3 = 0$)

Requires $\beta_3 = 0$ in the cubic model.

'linearHypothesis(mod, "X3 = 0")' $\implies F = 11.581, p = 0.00234 < 0.05$.

Conclusion: Reject H_0 . Total cost is not quadratic; cubic term X^3 is essential.

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Part (d): Linear Average Cost Test ($H_0 : \beta_0 = \beta_3 = 0$):

- **Average Cost Equation:** $Y_i/X_i = \beta_0/X_i + \beta_1 + \beta_2 X_i + \beta_3 X_i^2 + \varepsilon_i/X_i$.
- Linear AC ($Y_i/X_i = \beta_1 + \beta_2 X_i + u_i$) requires $\beta_0 = \beta_3 = 0$.
- **R Output:** 'linearHypothesis(mod, c("(Intercept)=0", "X3=0"))'

$F = 93.90, \quad p\text{-value} = 4.48 \times 10^{-12} < 0.05 \implies \text{Reject } H_0.$

Question 2(e)-(g): Ramsey RESET Specification Test

Ramsey RESET Principle

Tests for functional form mis-specification / omitted non-linear terms by adding powers of fitted values ($\hat{Y}^2$, $\hat{Y}^3$) to the model ('lmtest::resettest()').

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- **Cubic Cost Model ($Y \sim X + X^2 + X^3$):**
 - With $\hat{Y}^2$: RESET = 0.987, $p = 0.3307 > 0.05$
 - With $\hat{Y}^2$, $\hat{Y}^3$: RESET = 0.546, $p = 0.5871 > 0.05 \implies$ **Passes RESET!**
- **Log-Log Cost Model ($\ln Y \sim \ln X$):**
 - With $\hat{Y}^2$: RESET = 26.229, $p = 0.000027 < 0.05$
 - With $\hat{Y}^2$, $\hat{Y}^3$: RESET = 32.752, $p = 1.38 \times 10^{-7} < 0.05 \implies$ **Fails RESET!**

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Conclusion (g)

The RESET strongly supports the **cubic cost function** as correctly specified, while rejecting the log-log specification.

Exam Practice Question (MCQ)

Question Setup

Fit $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$ where X_{1i}, X_{2i} are both binary (0 or 1). Assume Y_i truly depends only on X_1 and X_2 . The model fails RESET ($p < 0.05$) using $\hat{Y}^2$. What omitted term does this suggest?

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Detailed Logical Analysis:

- For binary variables, $0^2 = 0$ and $1^2 = 1 \implies X_{1i}^2 = X_{1i}$ and $X_{2i}^2 = X_{2i}$.
- Thus, squared terms X_{1i}^2 and X_{2i}^2 are already implicitly included!
- Expanding $\hat{Y}_i^2 = (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i})^2$ introduces the **cross-product term** $X_{1i}X_{2i}$.
- A failing RESET indicates an omitted interaction term $X_{1i}X_{2i}$.

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Correct Answer

Option (1) cross product only ($X_{1i}X_{2i}$)

MCQ Extension: What if X_1, X_2 are Non-Binary?

Comparison: Binary vs. Continuous Regressors

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Key Distinction:

- **Binary Variables (0 or 1):** $X_{1i}^2 = X_{1i}$ and $X_{2i}^2 = X_{2i}$. Squared terms add no new information. $\hat{Y}^2$ proxies **only** for the cross-product $X_{1i}X_{2i} \implies$ **Option (1)**.
- **Continuous (Non-Binary) Variables:** $X_{1i}^2 \neq X_{1i}$ and $X_{2i}^2 \neq X_{2i}$. Squared terms X_{1i}^2, X_{2i}^2 are non-linear and **not** in the linear model.
 $\hat{Y}^2 = (\hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i})^2$ proxies for **both squared terms and cross-product** ($X_{1i}^2, X_{2i}^2, X_{1i}X_{2i}$).

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 $\hat{Y}^2 = (\hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i})^2$ proxies for **both squared terms and cross-product** ($X_{1i}^2, X_{2i}^2, X_{1i}X_{2i}$).

Answer for Non-Binary (Continuous) Regressors

If X_{1i}, X_{2i} are non-binary, the answer changes to **Option (3) squares and cross product ($X_{1i}^2, X_{2i}^2, X_{1i}X_{2i}$)!**